

Skills Summary

Random Samples and Simple Probability Models

Use this reference while completing your worksheet or when studying.

Two ideas run through everything here. A *probability model* says how likely an outcome is before you collect data; a *random sample* lets a small group speak for a large one. Both are fractions: part over whole.

Probability of one event

Rule: When the outcomes are equally likely, $P(\text{event}) = \frac{\text{favourable outcomes}}{\text{possible outcomes}}$, written in lowest terms.

Every probability lands between 0 and 1. To predict a count in many trials, multiply: predicted count = $P \times$ number of trials. The complement rule saves work: $P(\text{not } A) = 1 - P(A)$.

Example: A bag holds 8 marbles, all the same size: 3 are red. One is drawn without looking. Find $P(\text{red})$.

Step 1. The marbles are identical apart from colour, so the 8 outcomes are equally likely and the fraction model applies.

Step 2. Favourable is the 3 red marbles, possible is all 8 marbles: $\frac{3}{8}$, which has no common factor to cancel.

$$P(\text{red}) = \frac{3}{8}$$

Try it: A spinner has 12 equal sections and 5 of them are yellow. Find $P(\text{yellow})$.

check: $\frac{5}{12}$

Estimating from a random sample

Rule: In a random sample, the sample fraction estimates the population fraction. So divide the count in the sample by the sample size, then multiply by the population size: that product estimates the count in the whole population.

This only works when the sample is *random* — every member of the population having the same chance of being chosen. A sample gathered where the answer is more common (a skate park, a team practice) is biased, and no arithmetic can repair it.

Example: In a random sample of 25 of a town's 400 dogs, 9 are brown. Estimate how many of the town's dogs are brown.

Step 1. The sample was random, so the brown fraction in the sample is the best estimate of the brown fraction in the whole town — scale it up.

Step 2. $\frac{9}{25}$ of 400 is $\frac{9}{25} \times 400 = 9 \times 16$.

About **144** brown dogs

Try it: In a random sample of 20 of a school's 300 students, 7 ride the bus. Estimate how many students ride the bus.

check: 105

Summarizing the sample you collected

Rule: Order the values first. The **median** is the middle value (the average of the two middle values when the count is even). The **mean** is the total divided by how many values there are.

Split the ordered list at the median: Q_1 is the median of the lower half, Q_3 the median of the upper half, and the **interquartile range** is $Q_3 - Q_1$, the spread of the middle half. The **range**, largest minus smallest, is a different number and uses only the two extremes.

Example: Sample: 4, 6, 9, 10, 12, 15, 21. Find the median and the interquartile range.

Step 1. Both measures need the values in order and the list split at the middle, so locate the median first and work outward.

Step 2. Seven values, so the median is the fourth. Lower half 4, 6, 9 gives $Q_1 = 6$; upper half 12, 15, 21 gives $Q_3 = 15$; then $15 - 6$.

Median **10**, IQR **9**

Try it: Sample: 2, 5, 7, 8, 11, 13, 16. Find the median and the interquartile range.

check: median 8, IQR 8